

$$x(z) = \sum_{n=0}^{\infty} n 2^n z^{-n} = \sum_{n=0}^{\infty} n \left(\frac{2}{z}\right)^n$$

$$\frac{2}{z} = r$$

$$\rightarrow x^+(z) = \sum_{n=0}^{\infty} n r^n$$

$$S = 0 + r + 2r^2 + 3r^3 + \dots$$

$$rS = r^2 + 2r^3 + 3r^4 + \dots$$

$$S(1-r) = r + r^2 + r^3 + \dots$$

$$S(1-r) = \frac{r}{1-r}$$

$$S = \frac{r}{(1-r)^2}$$

$$\rightarrow x^+(z) = \frac{2}{\left(1 - \frac{2}{z}\right)^2}$$

$$x^+(z) = \frac{2z}{(z-2)^2}$$

$$x^+(z) = \frac{2z}{(z-2)^2}$$

~~$$x(n+1)$$~~

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$x(n+1)u(n+1) - 2x(n)u(n)$$

$$(n+1)2^{n+1}u(n) \quad n \geq 0$$

↓

same as $n \geq 1$

since first term is 0

$$= \frac{2z}{(z-2)^2}$$

$$x(n+1) \leftrightarrow X^+(z) = \sum_{n=0}^{\infty} x(n+1) z^{-n}$$

$$= \sum_{n=0}^{\infty} x(n+1) z^{-n+1} + z$$

$$= z \left[X^+(z) - x(1)z^{-1} \right]$$

$$\Rightarrow z X^+(z) - \frac{x(1)}{z}$$

$$T(n) = x(n+1)u(n+1)$$

$$\rightarrow T(0) = x(1)u(1)$$

$$\rightarrow x(1) = 1$$

$$\Rightarrow \text{Z transform of } x(n+1) = \cancel{z X^+(z)}$$

$$= z X^+(z) - z^{-1}$$

$$\Rightarrow \frac{d}{dz} (z x^+(z) - z^{-1}) - 2 (x^+(z)) = \frac{2z}{(z-2)^2}$$

$$\Rightarrow \mathbb{Z} (z x^+(z) - z^{-1}) - 2 (x^+(z)) = \frac{2z}{(z-2)^2}$$

$$\Rightarrow (z-2) x^+(z) = \frac{2z}{(z-2)^2} + z^{-1}$$

$$x^+(z) = \left[\frac{2z}{(z-2)^2} + \frac{1}{z} \right] \frac{1}{(z-2)}$$

$$x^+(z) = \frac{2z}{(z-2)^3} + \frac{2z^2 + z^2 + 4 - 4z}{z(z-2)^3}$$

$$x^+(z) = \frac{3z^2 - 4z + 4}{z(z-2)^3}$$

~~$$x^+(z)$$~~